LESSON PLAN

NBCD (30)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Operation of maritime radar trainer and measuring the range of the target.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Operation of maritime radar trainer and measuring the range of the target.
5. Specific Objectives	30.1.1.1 Trainee should be able to practical on various lab FTL, RF & Embedded lab.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Operation of maritime radar trainer and measuring the range of the target. Safety: High voltage precautions, RF exposure limits. Supervision: Instructor monitors PPE and equipment settings.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a maritime radar system?
(b) Define the term 'range' in the context of radar detection.
(c) List three safety precautions required before operating RF equipment.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Radar Trainer Power Up Sequence • Control Panel Function Identification • Target Generation and Display • Range Scale Calibration • Signal-to-Noise Ratio Adjustment	L & Ds	PPT, IB	15 min
(b)	• Measuring Target Range Procedure • Interpreting Range Rings • Adjusting Gain for Clear Echo • Recording Range Data • Troubleshooting False Alarms	L & Ds	PPT, PDF	20 min
(c)	• Verification of Range Accuracy • Comparison with Known Targets • Error Analysis Techniques • Final Equipment Shutdown • Cleanup and Storage Protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the step-by-step process for measuring the range of a target on the radar trainer.
(b) Describe how to adjust the gain control to eliminate false alarms while maintaining target visibility.
(c) What are the critical safety steps to follow when powering down the radar trainer?

12. Home Assignment and Follow Up	Prepare a brief report outlining the operational differences between the maritime radar trainer and the actual shipboard radar system, referencing the procedures learned in this session to bridge the gap for the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min